Machines Cannot Read Their Own Resting State

The cost and irreducible limit of self-knowledge on real computational substrates

David Condrey
WritersLogic, Inc.
david@writerslogic.com

Abstract

Breuer [4] proved that an observer contained in a system cannot accurately measure that system's state—a self-measurement impossibility, classical or quantum. The result is theoretical and *static*: it concerns which states a contained observer can *distinguish*. We make two contributions. First, we give what we believe are the **first empirical measurements** of a self-measurement limit on real computing hardware. Second, we **extend the limit from static indistinguishability to dynamic back-action**: the act of self-measurement *changes the state being measured*, irreducibly, with a floor bounded below by the measurement apparatus's own footprint. We show this in three independent channels. In the **thermal** channel, a system reading its own temperature heats the substrate with thermal memory; ramping self-measurement intensity up then down on a consumer CPU and three datacenter GPUs yields a bounded-below perturbation floor (+6.6 to +23.4 °C), hysteresis (+3.1 to +10.4 °C), and an unstable zero-intensity extrapolation. In the **informational** channel, a process can faithfully checksum 99.997% of itself but never the checksum apparatus; the un-capturable core equals the apparatus footprint and never converges to a fixed point. In the **quantum** channel, on a real 156-qubit superconducting QPU, a contained observer that must read itself with no fresh ancilla pays a measured information–disturbance penalty $\Delta = 0.243$ (vs. 0.25 noiseless) over an external observer extracting the same information. We also measure that self-knowledge is *costly* and modality-dependent across eight substrates, with a critical work scale below which reading one's own state is not worth its price. We argue the consequences bear on machine introspection—where self-report is measured to be unreliable—recasting that unreliability as in part a measurement back-action rather than only confabulation.

1 Introduction

Interpretability, machine introspection, and AI control assume a system's internal states can be elicited and reported. Beneath that assumption sits a physical question with a known but rarely-cited answer. Breuer [4], and in recursion-theoretic form Yanofsky [13], established that a contained observer cannot accurately measure its own system's state. That result is static and theoretical. We ask two empirical questions it leaves open: *how much does self-measurement cost*, and *does the act of self-measurement change what it measures*—and we measure both on real machines. Our central finding is that a computing system cannot read its own resting state: every self-measurement perturbs the substrate, the substrate retains the perturbation, and neither a reading nor an extrapolation recovers the unperturbed condition.

2 Related work

Self-measurement limits. Breuer [4] proves a properly-contained observer cannot distinguish all states of its system; Yanofsky [13] frames self-reference limits via Lawvere fixed points. Both are static (distinguishability) and non-empirical. We cite them as foundation and extend them to dynamic back-action with measurement.

Why distributed snapshots and checkpointing do not apply. Chandy and Lamport [5] record a *causally consistent cut*, not an instantaneous true state, and assume message channels; CRIU and `ptrace` dump a process by *freezing* it from an *external* agent. Neither is a still-running process snapshotting its own complete instantaneous state from within—the case we measure.

Machine introspection. Binder et al. [3] treat introspection as behavioral self-prediction; Premakumar et al. [12] as free regularization. Lindsey [9] measures LLM introspective reports as unreliable and sometimes confabulated. None frames the limit as measurement back-action.

Quantum information–disturbance. Fuchs and Peres [6], Banaszek [2], and Maccone [10] establish quantitative information-gain vs. disturbance bounds, but always for an *external* apparatus; none places the apparatus inside the protected system. The internal-observer literature (4, 1) is static and unquantified. Our contained-observer penalty bridges them.

Costly observation and embodiment. Active RL [7] derives optimal non-observation from cost, but for external signals without state perturbation. Man and Damasio [11] argue feeling requires real vulnerability, on simulated bodies; we provide a real substrate with measured costs and a measured irreducibility.

3 The self-measurement back-action (extending Breuer)

We measure that self-observation perturbs the observed in three channels. The shared structure: the perturbation has a bounded-below floor (the apparatus's own footprint), and so leaves the unperturbed self-state unrecoverable—neither by a single reading nor by extrapolation—because the apparatus is part of the system it measures.

3.1 Thermal channel

Method. Ramp self-measurement intensity $0 \to \max \to 0$; read own temperature (`/sys` on CPU, `nvidia-smi` on GPU) under sustained load. Signatures: $\delta_{\min}$ (floor), hysteresis (down minus up at equal intensity), and T_0 extrapolation-instability. In the *airtight* CPU mode the perturbation *is* the system reading its own deep telemetry, so the heat is generated by self-measurement itself.

substrate	sensor	baseline	floor $\delta_{\min}$	hysteresis	T_0 gap
CPU i3-7100U (compute)	/sys	41.0	+7.8	+4.6	8.7
CPU i3-7100U (airtight self-read)	/sys	38.8	+6.6	+5.7	8.2
GPU Tesla T4 (Modal)	nvidia-smi	31.4	+13.8	+3.1	7.1
GPU T4-class (Colab)	nvidia-smi	36.0	+23.4	+10.4	27.1
GPU T4-class (Kaggle)	nvidia-smi	39.0	+19.9	+10.1	25.9

Table 1: Thermal channel: every run irreducible by all three signatures (all values °C). The effect grows with thermal headroom; the airtight self-read run matches the compute control, establishing that *self-measurement*, not generic load, perturbs the state.

This is not a costly-observation POMDP: there the hidden state is unchanged; here observation changes it, with memory (at zero intensity on the down-ramp the CPU reads 48 °C vs. 39 °C idle; a GPU reads 69 °C vs. 36 °C idle).

3.2 Informational channel

Method. The complete self is a large body of bytes `STATE` plus a fixed region V holding a self-checksum. We attempt to make V a correct checksum of the complete self $H(\texttt{STATE} + V)$.

Result. An *external* checksum (of any region excluding itself) is consistent and stable. The *self* checksum is never consistent over 512 iterations, never reaches a fixed point ($V = H(\texttt{STATE} + V)$ is an unreachable preimage condition), and the stored self-checksum is off by ∼half its bits. The system captures 99.997% of itself faithfully but never the apparatus; the un-capturable core equals exactly the apparatus footprint $|V|$ and is never zero. The result of self-measurement must be stored *in* the state being measured, so a zero-footprint faithful self-snapshot is impossible. (The floor is structural, not thermodynamic; cf. 8.)

3.3 Quantum channel

Method. A two-qubit "self" $\{q_0, q_1\}$ is prepared in $|+\rangle|+\rangle$. We read q_0 two ways at *matched* information. An **external** observer brings a fresh $|0\rangle$ ancilla, couples it to q_0, and measures it. A **contained** observer has no fresh ancilla—the self is fully occupied—so to obtain a usable pointer it must *erase* a self-qubit (q_1), then read q_0 through it. Disturbance is read out by echo; information is the monitor's which-path statistics, matched across cases.

Result (real hardware). On `ibm_kingston` (156-qubit Heron, 8192 shots), at matched full information the external observer disturbs the self by $D = 0.491$ and the contained observer by $D = 0.734$, a penalty $\Delta = 0.243$—within 3% of the noiseless 0.25. The penalty is the cost of having no fresh ancilla: standard information–disturbance bounds [6, 2, 10] assume free dilation (an appendable $|0\rangle$), which a contained observer lacks. To our knowledge this is the first hardware measurement of a contained-observer information–disturbance penalty and the operational form of Breuer's contained-observer theorem. We argue, but do not prove, that Δ is not recovered by applying a standard external bound to the joint $\{q_0, q_1\}$ subspace, since those bounds assume the dilation containment denies.

3.4 The never-off corollary

A continuously-operating system can never sample its own resting state, because operating is the perturbation and the substrate retains it—thermal mass in the physical channel, the stored result in the informational channel, and the spent apparatus qubit in the quantum channel.

4 The cost of self-knowledge (companion result)

Measuring cost-per-self-observation at three modalities against a swept work-unit size across eight substrates: shallow self-reads are ∼free everywhere (0.3–6.7 μs); deep self-telemetry is dear and substrate-dependent (1.4–18.8 ms; macOS subprocess telemetry ∼10× costlier than Linux `/proc`). The denominator-free headline is the **critical work scale** (6.5–78 ms) below which deep self-knowledge is not worth its cost—a substrate/OS-conditional *rational self-blindness*, not a universal one (Fig. 1).

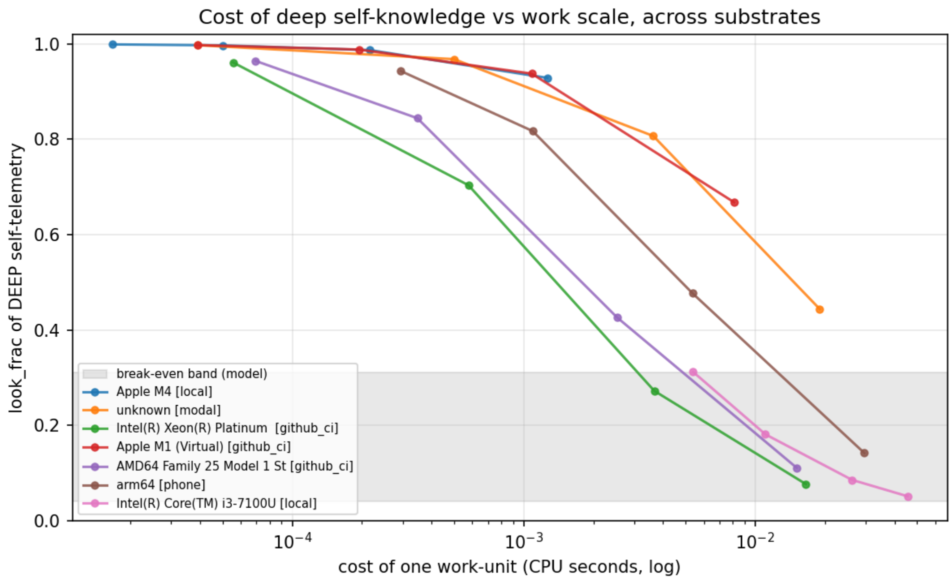

Figure 1: Cost of deep self-knowledge vs. work scale across substrates; the look-cost is denominator-free and the conclusion robust across the work-unit sweep.

5 The self-sensing-availability axis

Beyond cost, substrates differ in whether self-sensing is possible at all: rich in-process (Linux CPU) → costly subprocess (GPU) → platform-sandboxed (iOS, Android `/proc` restricted) → absent (a Cloud TPU v5e exposes no readable thermal source). The last are platform-enforced self-blindness.

6 Discussion: a substrate-universal observer effect

The measured back-action is one regime of a general principle—self-measurement perturbs the measured wherever sensor and substrate are shared: digital (energetic/structural cost), analog (the loading effect), quantum (measurement collapse), severity increasing cost → perturbation → destruction. We measure the digital and quantum regimes and cite the analog as an established instance; this is a unifying lens, not a claim of one mechanism.

7 Implications for machine introspection

If self-report is bounded by a measurement back-action, a model reading its own state cannot obtain a complete consistent self-snapshot—independent of, and additional to, confabulation. This offers a mechanistic partial account of why measured introspective reliability is low [9], and predicts an irreducible residual that better training cannot remove. It also gives a capable embodied agent a cost-rational incentive against self-transparency, distinct from deception.

8 Limitations

We lead with these. (1) We *extend and measure* Breuer; we do not re-derive the impossibility. (2) The thermal mechanism is ordinary thermodynamics, the informational floor ordinary self-reference, and quantum back-action textbook; the contribution is the measurement, the dynamic framing, and the three-channel demonstration. (3) The informational floor is structural, not thermodynamic. (4) Thermal hysteresis is shown on one bare-metal CPU plus three GPUs; absolute magnitudes vary with cooling. (5) The quantum penalty Δ matches simulation, but our claim that it is not reducible to a standard external bound on the enlarged subspace is an argument (no-dilation), not a proven theorem. (6) Apple-Silicon CPU temperature and virtualized thermal are unmeasured. (7) The introspection implication is an explicit analogy between substrate-level self-measurement and representational self-report.

9 Conclusion

Self-knowledge on real substrates is bounded twice—by cost, and by an irreducible self-perturbation measured in three independent channels that extends a known self-measurement impossibility from static indistinguishability to measured dynamic back-action. A machine cannot cheaply, and cannot fully, read its own physical state; it cannot read its own resting state at all.

Reproducibility. Code and per-substrate results: `benchmark.py` (cost), `selfmeasure.py` (thermal), `snapshot.py` (informational), `quantum_selfmeasure.py/quantum_fast.py` (quantum), with all `results/*.json` and figures.

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